

HUNAIN AHMED

AI / Machine Learning & Full-Stack Systems Engineer

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Karachi, Pakistan
Available for Engineering Roles

TECHNICAL SKILLS & CORE COMPETENCIES

CORE // STACK

Languages:	Python, TypeScript, JavaScript (ES6+), SQL, C/C++, HTML5/CSS3
AI & Machine Learning:	PyTorch, TensorFlow, Scikit-Learn, LLMs, Autonomous Agents, Multi-Modal Gemini API, Vector Embeddings, Hybrid RAG, NLP, Cosine Similarity, SVD
Full-Stack & Systems:	React 19 / 18, FastAPI, Node.js, Canvas 2D (60fps), Web Audio API, WebSockets, REST APIs, Tailwind CSS, Zustand, AsyncIO
Infrastructure & Tools:	PostgreSQL (pgvector), Redis, Docker, Git, GitHub Actions (CI/CD), Linux/Bash, Mapbox API, Vite

SOFTWARE ENGINEERING & AI SYSTEMS EXPERIENCE

SYSTEMS ARCHITECTURE

Klyptic — Browser-Native Non-Linear Video Editor | Lead Architect & Systems Developer 2025 – Present

- React 18, TypeScript, HTML5 Canvas 2D, Web Audio API, Tailwind CSS, Zustand
- Architected an in-browser non-linear editing suite delivering **zero-server video rendering** and sustained **60 FPS compositing** via hardware-accelerated Canvas 2D and requestAnimationFrame loops.
 - Engineered non-destructive multi-track sequencing with sub-frame keyframe interpolation, dynamic transition procedural shaders, and audio-video timeline synchronization.
 - Integrated local AI auto-captioning engine with millisecond word-timestamp alignment and offline audio graph rendering via Web Audio API context, reducing backend compute costs to zero.

Trade Bot Gemini — Autonomous Multi-Modal Trading Agent | AI Engineer & Creator 2025 – Present

- Python, Google Gemini Multi-Modal API, NumPy, Pandas, AsyncIO, Autonomous Reasoning Loops
- Engineered an autonomous multi-modal quantitative trading agent ingesting live price action, order-book depth, and macroeconomic sentiment to generate structured, risk-weighted hypotheses.
 - Designed a structured self-auditing reflection loop enforcing strict downside limits, executing decision inference cycles in **under 800ms** with zero policy breaches.
 - Implemented dynamic volatility calibration and adaptive position sizing based on real-time rolling drawdown thresholds.

Pakistan Super App — Mobility & Roadside Logistics Platform | Full-Stack Systems Engineer 2024 – 2025

- TypeScript, React Native, Node.js, WebSockets, Mapbox API, Redis, Monorepo
- Architected a cross-platform mobility monorepo unifying passenger, driver, and emergency roadside recovery mobile clients with a scalable Node.js backend service layer.
 - Implemented a **sub-second reverse-auction bidding protocol** over persistent bidirectional WebSockets with automated geospatial geohash proximity dispatch.
 - Engineered battery-optimized background GPS coordinate synchronization with automatic connection failure recovery.

Movie Recommender AI — Vector Similarity & Hybrid ML Engine | ML Developer 2024

- Python, Scikit-Learn, Pandas, NLP Embeddings, Truncated SVD, Cosine Similarity
- Formulated a hybrid recommendation engine synthesizing high-dimensional NLP plot embeddings with sparse collaborative rating matrices to mitigate user cold-start bias.
 - Optimized vector space retrieval using Truncated SVD dimensionality reduction, achieving **< 10ms retrieval latency** across large candidate catalogs.

COMPETITIVE PROGRAMMING & TECHNICAL HONORS

AWARDS & CONTESTS

PROCOT — National SQL & Database Optimization Contest | FAST-NUCES FAST-NUCES

Competed in intensive database query-solving; authored optimized multi-table joins, subqueries, and analytical SQL scripts under strict timed limits.

SuperComp — Speed Programming & Algorithmic Problem Solving | Bahria University (BUKC) BUKC

Solved complex algorithmic challenges and edge-case test suites in C++ and Python under strict timed conditions, optimizing time and space complexity.

TEKNOFEST International Aerospace & Technology Festival | Karachi Expo Center Expo Center

Participated in international technology competition tracks, collaborating on technical problem-solving and innovation-focused engineering solutions.

EDUCATION & FORMAL FOUNDATIONS

ACADEMIC FOUNDATION

Bahria University Karachi — Bachelor of Science in Artificial Intelligence (BS AI) 2023 – Present

Key Coursework: Deep Learning, Neural Networks, Natural Language Processing, Computer Vision, Probabilistic Graphical Models, Data Structures & Algorithms, Database Systems, Systems Architecture.